BeeQ: Final Supporting Information

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This supplement accompanies the full revision based on the reviewed manuscript and the two reviewer reports. Original results and the retrospective MLP analyses are separately identified. Tables D1–D7 document the fixed representation and original campaign; Tables M1–M7 provide the complete additional MLP analysis. Supporting numerical files are released at full precision; displayed values are rounded.

1 Fixed inputs and descriptor definitions

The reference campaign is `results/campaigns/BEEQ_FINAL_NESTED_STRUCT_IMPL_20260827T144143Z`. Development, historical holdout and CR8 use the official corrected inputs, not similarly named auxiliary copies. No descriptor, compound, endpoint, or partition is changed in this revision. The exact input hashes are in `data/SHA256SUMS.csv`; fit-level reproducibility uses these frozen values.

Table D1. The ordered X10 representation. RDKit version 2026.03.5.

Feature	Definition and reproducibility scope
MolLogP	RDKit Wildman–Crippen octanol/water logP estimate.
MolWt	RDKit molecular weight using standard atomic weights.
TPSA_SP	`rdMolDescriptors.CalcTPSA(mol, includeSandP=True)`.
NumHDonors	RDKit hydrogen-bond donor count.
NumRotatableBonds	RDKit rotatable-bond count with the frozen default definition.
NumAromaticRings	RDKit aromatic-ring count.
nHalogen	Atom count for atomic numbers 9, 17, 35, and 53 (F, Cl, Br, I).
n_OP	Number of matches of `[P;X4](=[O,S])([O,S][#6])([O,S][#6])`.
LiPHEX_prediction	Supplied frozen prediction of n-hexadecane/water logP from LiPHEX (main reference 18). No BeeQ toxicity labels are fitted by this revision.
sasa002_frac_polar_hetero_only	Supplied frozen three-dimensional fraction of accessible surface attributed to polar heteroatoms. A complete conformer/probe-radius generation recipe is not established by the retained modeling artifacts.

Eight RDKit-derived columns have a retained parity audit. The two supplied in-house columns are preserved, not regenerated. Reproducing model results from the supplied X10 tables is supported; regenerating the in-house descriptors from SMILES requires the originating workflows. Verified lineage begins at the frozen corrected 893-row corpus; a complete per-structure exclusion ledger from the 1,035-row source is not established by the retained modeling artifacts. This provenance boundary does not affect the paired class-weight comparison on fixed X10.

Table D2. Partition and outer-fold class distributions. Outer training complements contain only Development molecules.

Set	N	Non-toxic	Toxic	Toxic fraction	Clusters
BeeQ893	893	622	271	0.3035	462
Development	712	490	222	0.3118	374
Historical holdout	181	132	49	0.2707	88
CR8	8	6	2	0.2500	—
Outer validation 1	137	97	40	0.2920	—
Outer validation 2	105	68	37	0.3524	—
Outer validation 3	147	110	37	0.2517	—
Outer validation 4	181	122	59	0.3260	—
Outer validation 5	142	93	49	0.3451	—

Exact outer/inner/final training counts and memberships for the MLP analysis are in `SPLIT_COUNTS.csv` and `SPLIT_MEMBERSHIP.csv` within the additional bundle. All outer and inner evaluation cluster intersections are

zero. The MLP's native early-stopping reservation is a different, molecule-level split and shares clusters with gradient training; see Tables M3 and M7. Cluster membership separation does not guarantee that every pair in different groups is below a Tanimoto threshold.

2 Original selection protocol and results

Search spaces below are from the final campaign, not the older exploratory runner. Every candidate is evaluated using the same four inner cluster folds for a given outer fold. Selection uses mean unweighted inner AUROC. Final selection instead uses the five frozen Development folds and is a selection diagnostic, not nested performance. Thresholds use that configuration's OOF scores, maximum MCC, maximum BA, minimum absolute threshold, then minimum numerical threshold. LR/RF/MLP scores are probabilities; SVC scores are decision values, so threshold magnitudes are not directly comparable.

Table D3. Original hyperparameter grids and fixed settings.

Family	Search space / fixed settings
LR	$C = \{0.01, 0.1, 1, 10, 100\}$; liblinear, max_iter=5000, balanced class weights; training-fitted StandardScaler.
RF	max_depth $\in \{\text{None}, 6\}$; min_samples_leaf $\in \{1, 3\}$; 300 trees, sqrt features, balanced_subsample, n_jobs=1.
Original MLP	Hidden layers $\{(32), (32, 16)\}$; alpha $\{0.0001, 0.001, 0.01\}$; initial rate $\{0.0003, 0.001\}$ (12 candidates). ReLU, Adam, max_iter=2000, batch size auto, early stopping, validation_fraction=0.1, tol=10^{-4}, n_iter_no_change=10, no weights; training-fitted StandardScaler.
RBF	$C = \{0.1, 1, 10, 100\}$; $\gamma = \{0.01, 0.03, 0.1, 0.3\}$; balanced precomputed SVC.
Product QK	Same C grid; scale $s = \{0.125, 0.25, 0.5, 1\}$; balanced precomputed SVC, exact separable rotations.
IQP-ZZ QK	Same C and scale grids; interaction strength 1; balanced precomputed SVC, exact diagonal nearest-neighbor phases.

Table D4. Final original configurations and frozen thresholds.

Model	Selected configuration	Threshold
Logistic Regression	{'C': 100.0}	0.660538
Random Forest	{'max_depth': None, 'min_samples_leaf': 3}	0.682592
MLP original	{'alpha': 0.01, 'hidden_layer_sizes': [32, 16], 'learning_rate_init': 0.0003}	0.524583
Matched RBF	{'C': 1.0, 'gamma': 0.03}	0.915655
Product QK	{'C': 1.0, 'feature_scale': 0.25}	0.646198
IQP-ZZ QK	{'C': 1.0, 'feature_scale': 0.125, 'interaction_strength': 1.0}	0.383521

Table D5. Original fixed-configuration label-permutation control, 200 permutations. Real scores are pooled nested OOF scores; the null fits use final configurations. This is an incomplete control for selection optimism, not a nested-HPO permutation test. Zero exceedances are a count, not a zero p-value.

Model	Real OOF ROC	Perm. mean	Perm. SD	Perm. P95	Count $\geq$ real
Logistic Regression	0.7252	0.4996	0.0310	0.5516	0/200
Random Forest	0.7565	0.4958	0.0301	0.5430	0/200
MLP original	0.6677	0.5013	0.0256	0.5419	0/200
Matched RBF	0.7154	0.4978	0.0304	0.5471	0/200
Product QK	0.7171	0.4980	0.0304	0.5514	0/200
IQP-ZZ QK	0.7140	0.4979	0.0303	0.5493	0/200

3 Kernel geometry reconstruction and paired statistics

The exact Product expression and canonical NumPy feature maps were re-evaluated without labels or model fitting on the 712 Development rows. The final supervised-selected parameters are RBF gamma=0.03, Product scale=0.25, IQP-ZZ scale=0.125 and coupling=1. The separate RBF control uses gamma=$0.25^2/4 = 0.015625$. Entropy-based effective rank is $\exp(-\sum_j p_j \log p_j)$, where p_j normalizes nonnegative eigenvalues of the *uncentered* Gram matrix. CKA uses double-centered matrices and their normalized Frobenius inner product.

Table D6a. Geometric diagnostics. Off-diagonal statistics use all 253,116 unique pairs.

Kernel	Effective rank	Off-diag. mean	Off-diag. SD
Matched RBF	5.517	0.609	0.210
Product QK	3.086	0.753	0.165
IQP-ZZ QK	3.156	0.750	0.167
Width-control RBF	3.078	0.757	0.160

Table D6b. Reconstructed centered kernel alignments.

Pair	CKA
Product – selected RBF	0.982454
IQP-ZZ – selected RBF	0.985072
Product – IQP-ZZ	0.999506
Product – width-control RBF	0.999794
IQP-ZZ – width-control RBF	0.999312
Selected RBF – width-control RBF	0.983587

The Product analytic/simulated maximum difference is 3.89×10^{-15}. These diagnostics reproduce the bandwidth interpretation and correct the reviewed abstract's inter-quantum CKA. They do not estimate the geometric-difference bound, exponential concentration with qubit count, trainability, or finite-shot costs. The script and complete diagnostics are in `manuscript/final_revision/build_assets.py` and `audit/KERNEL_GEOMETRY.json`. An approximate power-law exponent is omitted from the manuscript because its value depends on the fitting criterion; the defined log-space through-origin calculation is recorded in the JSON for traceability.

Table D6c. Original paired cluster-bootstrap differences, conditional on pooled OOF predictions. 2,000 Butina-cluster resamples, seed 20260824. These are descriptive intervals without refitting, selection uncertainty, or multiplicity adjustment.

Comparison	Metric	Pooled delta	Lower 95%	Upper 95%
Random Forest - Logistic Regression	AUROC	0.0313	-0.0210	0.0932
Random Forest - Logistic Regression	AUPRC	0.0295	-0.0235	0.0812
Random Forest - MLP original	AUROC	0.0889	0.0217	0.1616
Random Forest - MLP original	AUPRC	0.0986	0.0366	0.1522
Product QK - Matched RBF	AUROC	0.0017	-0.0197	0.0242
Product QK - Matched RBF	AUPRC	-0.0281	-0.0536	0.0008
IQP-ZZ QK - Matched RBF	AUROC	-0.0014	-0.0237	0.0255
IQP-ZZ QK - Matched RBF	AUPRC	-0.0109	-0.0380	0.0210
Product QK - IQP-ZZ QK	AUROC	0.0032	-0.0209	0.0279
Product QK - IQP-ZZ QK	AUPRC	-0.0172	-0.0591	0.0206

Table D7a. Original AUROC per frozen outer fold.

Model	F1	F2	F3	F4	F5
Random Forest	0.6879	0.7476	0.7197	0.7433	0.8416
Logistic Regression	0.5716	0.6904	0.8000	0.7726	0.8293
Product QK	0.5619	0.6590	0.8079	0.7833	0.7961
IQP-ZZ QK	0.5802	0.6475	0.8285	0.7002	0.8086
Matched RBF	0.5588	0.6419	0.7813	0.7470	0.8240
MLP original	0.4889	0.6367	0.7921	0.6826	0.7852

Table D7b. Original AUPRC (average precision) per frozen outer fold.

Model	F1	F2	F3	F4	F5
Random Forest	0.5270	0.6546	0.6193	0.6967	0.7889
Logistic Regression	0.4702	0.6125	0.6933	0.7311	0.7908
Product QK	0.4425	0.5547	0.7119	0.7443	0.6864
IQP-ZZ QK	0.5017	0.5707	0.7409	0.6465	0.7677
Matched RBF	0.4967	0.5604	0.7000	0.6960	0.7491
MLP original	0.4055	0.5206	0.6586	0.6281	0.7678

Original molecule-level OOF, holdout and CR8 predictions, thresholds, all metrics and original bootstrap/Y-randomization records remain in the reference campaign. CR8 identities and applicability flags use `06_EXTERNAL_CR8/CR8_DUAL_AD.csv`. Domain thresholds and Development leave-one-out measures were independently reconstructed in this revision; `audit/AD_RECONSTRUCTION.json` records agreement and Spearman correlation. No new predictor or representation is trained for these editorial checks.

Additional retrospective MLP analysis

The following sections document the validated additional sensitivity analysis in Tables M1–M7. They support Methods II-D/II-E and Results III-B/III-C/III-E. The original six-family results and the original standalone MLP supplement remain archived separately.

4 MLP Methods

The original MLP baseline used StandardScaler followed by scikit-learn 1.9.0 MLPClassifier, with ReLU activation, Adam, max_iter=2000, early_stopping=True, validation_fraction=0.1, and random_state=20260824. It was trained without class balancing. The implementation omitted sample weights when the MLP was introduced; no contemporaneous rationale establishes why. The recorded library supported sample_weight, so a library restriction cannot be cited as the explanation. LR, RF and the three SVC-based baselines retained their original weighting mechanisms. This initial asymmetry remains a limitation of the original comparison.

We conducted an additional retrospective sensitivity analysis using the same X10 features, molecules and frozen partitions. For every training fit, class weights were calculated as w_c=N_fit/(2 n_c,fit) from that fit's labels and passed only to the MLP as sample weights. The scaler and all evaluation metrics remained unweighted; no resampling was used. Analysis A fixed each outer fold's hyperparameters and the final configuration to those selected by the original unweighted MLP. Analysis B repeated the full 12-candidate search over hidden_layer_sizes={(32,), (32,16)}, alpha={0.0001,0.001,0.01}, and learning_rate_init={0.0003,0.001}, using five frozen outer folds and four inner cluster-disjoint StratifiedGroupKFold partitions with seeds 20260824+outer_fold. Mean unweighted inner AUROC selected the first highest-scoring ParameterGrid candidate.

For each variant and outer training partition, thresholds were estimated separately from complete inner OOF predictions by maximizing MCC, then balanced accuracy, then minimizing the absolute threshold and finally the threshold itself. Final configuration selection used the five frozen Development folds; its OOF predictions determined a final threshold under the same rule, followed by training on all Development. Selection OOF metrics are not an additional unbiased estimate of generalization. All configurations, thresholds and model artifacts were frozen before the new input CSV evaluations of the historical holdout and CR8. These datasets and historical predictions had already been observed, so the extension is retrospective, not a new blind evaluation.

Native early stopping was preserved in both variants. Its stratified 10% molecule reservation is not cluster disjoint, and StandardScaler is fitted before that reservation. Weights are derived from the enclosing CV training fit, including this native stopping reservation; no CV evaluation, holdout or CR8 labels contribute to the class counts. The native implementation carries these weights into stopping accuracy and normalizes both the loss gradient and L2 penalty by each minibatch's weight sum. Therefore Analysis A measures the effect of native weighted training, including its stopping behavior, rather than a pure loss-gradient effect with an invariant stopping trajectory. Architecture, optimizer, alpha, batch size, tolerance and stopping settings were not changed.

Differences were paired on the same outer folds. For comparison with the original project statistics, we also used 2,000 paired Butina-cluster bootstrap resamples of fixed pooled OOF scores (seed 20260824). These descriptive intervals condition on the fitted predictions, do not repeat training or selection, and do not constitute multiplicity-adjusted hypothesis tests. Cross-fold score calibration and overlapping training sets limit interpretation of pooled statistics and five-fold consistency.

5 Results

The original MLP was reproduced before interpreting balancing: all five nested OOF folds, selected configurations, thresholds, historical-holdout predictions and CR8 predictions agreed with the recorded campaign (maximum absolute score difference 1.89e-15; tolerance 1e-10). Its Development outer-fold AUROC was 0.677 ± 0.124 and AUPRC 0.596 ± 0.138; historical-holdout AUROC was 0.462. Thus the reviewed approximate values refer to the outer-fold mean and holdout, respectively; pooled nested OOF AUROC was 0.668.

With original hyperparameters fixed (Analysis A), outer-fold AUROC became 0.688 ± 0.100 and AUPRC 0.609 ± 0.119. With full retuning (Analysis B), AUROC was 0.679 ± 0.100 and AUPRC 0.594 ± 0.112. Paired mean AUROC changes were +0.0113 and +0.0021, respectively. At the fold-specific MCC thresholds, pooled toxic recall changed from 75/222 to 79/222 and 75/222, while false positives increased from 15/490 to 39/490 and 36/490. Recall improved in four folds but fell from 12/37 to 1/37 in the third fold in both balanced variants. Consequently sensitivity did not improve consistently. Pooled MCC decreased from 0.428 to 0.344 and 0.338.

The descriptive pooled AUROC bootstrap intervals included zero for both balanced-minus-original comparisons (A: -0.0242 to 0.0356; B: -0.0369 to 0.0258). For retuned pooled AUPRC the difference was -0.0474, with a descriptive interval of -0.0989 to -0.0009. This negative pooled result differs from the small outer-fold mean change and must not be described as a global discrimination improvement. No causal explanation of the MLP's overall inferiority is established by these results.

On the historical holdout, AUROC/AUPRC changed from 0.462/0.302 to 0.481/0.332 in A and 0.482/0.329 in B. Using separately frozen thresholds (original 0.524583, A 0.536224, B 0.528334), toxic recovery increased from 4/49 to 11/49 and 13/49, while specificity fell from 128/132 (0.970) to 116/132 (0.879) and 110/132 (0.833). False positives rose from 4 to 16 and 22. Discrimination therefore remained weak, whereas the major practical change was increased positive decisions with a specificity cost.

In CR8, both balanced variants detected Isocycloseram, which the original missed; all variants missed Ethiprole. Fluoxapiprolin became an additional false positive, while Florylpicoxamid remained a false positive. AUROC/AUPRC decreased from 0.833/0.583 to 0.667/0.450 in A and 0.583/0.367 in B. Recovering one of two toxic compounds cannot establish population sensitivity or robust regional generalization.

All 693 fits stopped before 2,000 iterations (12-77 iterations) without recorded training warnings. This verifies finite training and the configured stopping behavior, not optimization to a global or stationary solution. Native stopping reservations shared 14-32 clusters with their gradient-training subsets. An independent complete rerun reproduced all tabular outputs exactly. The balanced retuned MLP remains numerically below every other BeeQ model in mean outer AUROC; RF remains highest at 0.748. The quantum-versus-matched-RBF results are unaffected. The findings therefore preserve BeeQ's central comparative conclusions while qualifying the original MLP limitation: balancing improves retrospective toxic recovery at the price of more false positives and does not resolve its weak discrimination.

6 Development summary

Table M1a: AUROC/AUPRC: outer-fold mean and sample SD. BA/MCC: pooled nested OOF decisions. A fixes original hyperparameters; B repeats the search.

Variant	AUROC mean/SD	AUPRC mean/SD	Pooled BA	Pooled MCC
Original	0.6771 ± 0.1245	0.5961 ± 0.1383	0.654	0.428
Balanced A	0.6884 ± 0.1005	0.6086 ± 0.1186	0.638	0.344
Balanced B	0.6792 ± 0.0997	0.5942 ± 0.1119	0.632	0.338

Table M1b: Pooled nested Development decisions (490 negative, 222 toxic).

Variant	Sens.	Spec.	TP	FN	TN	FP
Original	0.338	0.969	75	147	475	15
Balanced A	0.356	0.920	79	143	451	39
Balanced B	0.338	0.927	75	147	454	36

7 External evaluation and frozen thresholds

Table M2a: historical_holdout: discrimination and frozen-threshold metrics.

Variant	AUROC	AUPRC	BALANCED_ACCURACY	MCC	THRESHOLD
Original	0.462	0.302	0.526	0.111	0.524583
Balanced A	0.481	0.332	0.552	0.129	0.536224
Balanced B	0.482	0.329	0.549	0.111	0.528334

Table M2b: historical_holdout: toxic recovery and specificity.

Variant	SENSITIVITY	SPECIFICITY	TP	FN	TN	FP
Original	0.082	0.970	4	45	128	4
Balanced A	0.224	0.879	11	38	116	16
Balanced B	0.265	0.833	13	36	110	22

Table M2c: CR8: discrimination and frozen-threshold metrics.

Variant	AUROC	AUPRC	BALANCED_ACCURACY	MCC	THRESHOLD
Original	0.833	0.583	0.417	-0.218	0.524583
Balanced A	0.667	0.450	0.583	0.149	0.536224
Balanced B	0.583	0.367	0.583	0.149	0.528334

Table M2d: CR8: toxic recovery and specificity.

Variant	SENSITIVITY	SPECIFICITY	TP	FN	TN	FP
Original	0.000	0.833	0	2	5	1
Balanced A	0.500	0.667	1	1	4	2
Balanced B	0.500	0.667	1	1	4	2

8 Outer folds and selected hyperparameters

Table M3a: Original: outer-fold metrics.

Fold	AUROC	AUPRC	BA	MCC	Sens.	Spec.	THRESHOLD
1	0.489	0.405	0.585	0.283	0.200	0.969	0.524038
2	0.637	0.521	0.625	0.325	0.324	0.926	0.539855
3	0.792	0.659	0.658	0.482	0.324	0.991	0.520559
4	0.683	0.628	0.619	0.416	0.237	1.000	0.527747
5	0.785	0.768	0.764	0.582	0.592	0.935	0.513438

Table M3b: Original: outer-fold class counts and decisions.

Fold	NEGATIVE	POSITIVE	TP	FN	TN	FP
1	97	40	8	32	94	3
2	68	37	12	25	63	5
3	110	37	12	25	109	1
4	122	59	14	45	122	0
5	93	49	29	20	87	6

Table M3c: Balanced A: outer-fold metrics.

Fold	AUROC	AUPRC	BA	MCC	Sens.	Spec.	THRESHOLD
1	0.536	0.450	0.552	0.113	0.300	0.804	0.561234
2	0.645	0.532	0.639	0.352	0.351	0.926	0.636153
3	0.781	0.631	0.514	0.143	0.027	1.000	0.650128
4	0.713	0.680	0.662	0.420	0.373	0.951	0.555593
5	0.767	0.749	0.768	0.566	0.633	0.903	0.541755

Table M3d: Balanced A: outer-fold class counts and decisions.

Fold	NEGATIVE	POSITIVE	TP	FN	TN	FP
1	97	40	12	28	78	19
2	68	37	13	24	63	5
3	110	37	1	36	110	0
4	122	59	22	37	116	6
5	93	49	31	18	84	9

Table M3e: Balanced B: outer-fold metrics.

Fold	AUROC	AUPRC	BA	MCC	Sens.	Spec.	THRESHOLD
1	0.536	0.450	0.552	0.113	0.300	0.804	0.561234
2	0.645	0.532	0.639	0.352	0.351	0.926	0.636153
3	0.781	0.631	0.514	0.143	0.027	1.000	0.650297
4	0.668	0.608	0.640	0.411	0.305	0.975	0.560083
5	0.767	0.749	0.768	0.566	0.633	0.903	0.541755

Table M3f: Balanced B: outer-fold class counts and decisions.

Fold	NEGATIVE	POSITIVE	TP	FN	TN	FP
1	97	40	12	28	78	19
2	68	37	13	24	63	5
3	110	37	1	36	110	0
4	122	59	18	41	119	3
5	93	49	31	18	84	9

Table M4: Selected outer-fold configurations and thresholds.

Variant	Fold	Layers	Alpha	Rate	THRESHOLD
Original	1	[32, 16]	0.0100	0.0010	0.524038
Original	2	[32, 16]	0.0100	0.0010	0.539855
Original	3	[32, 16]	0.0010	0.0010	0.520559
Original	4	[32, 16]	0.0100	0.0010	0.527747
Original	5	[32, 16]	0.0001	0.0010	0.513438
Balanced A	1	[32, 16]	0.0100	0.0010	0.561234
Balanced A	2	[32, 16]	0.0100	0.0010	0.636153
Balanced A	3	[32, 16]	0.0010	0.0010	0.650128
Balanced A	4	[32, 16]	0.0100	0.0010	0.555593
Balanced A	5	[32, 16]	0.0001	0.0010	0.541755
Balanced B	1	[32, 16]	0.0100	0.0010	0.561234
Balanced B	2	[32, 16]	0.0100	0.0010	0.636153
Balanced B	3	[32, 16]	0.0100	0.0010	0.650297
Balanced B	4	[32, 16]	0.0001	0.0003	0.560083
Balanced B	5	[32, 16]	0.0001	0.0010	0.541755

Original and A final configuration: layers (32,16), $\alpha = 0.01$, initial learning rate 0.0003. B final configuration: layers (32,16), $\alpha = 0.001$, initial learning rate 0.001. Final selection uses the five frozen Development folds, not external labels.

9 CR8 compound predictions

Table M5a: CR8 scores (positive class = toxic).

Compound	Truth	Original score	Balanced A score	Balanced B score
Isotianil	0	0.428	0.477	0.480
Fluoxapiprolin	0	0.487	0.573	0.575
Ethiprole	1	0.489	0.519	0.520
Cyclobutrifluram	0	0.471	0.520	0.522
Isoprothiolane	0	0.373	0.451	0.454
Isocycloseram	1	0.511	0.576	0.573
Spiropidion	0	0.415	0.466	0.468
Florylpicoxamid	0	0.535	0.645	0.653

Table M5b: CR8 decisions at separately frozen thresholds; only two compounds are toxic.

Compound	Truth	Original pred	Balanced A pred	Balanced B pred
Isotianil	0	0	0	0
Fluoxapiprolin	0	0	1	1
Ethiprole	1	0	0	0
Cyclobutrifluram	0	0	0	0
Isoprothiolane	0	0	0	0
Isocycloseram	1	0	1	1
Spiropidion	0	0	0	0
Florylpicoxamid	0	1	1	1

10 Paired cluster bootstrap

Table M6: 2,000 paired cluster resamples of pooled OOF predictions. Intervals are descriptive; the last column is not a p-value.

Comparison	METRIC	Delta	CI low	CI high	P_DELTA_GT_0
Balanced A - Original	AUROC	0.005	-0.024	0.036	0.621
Balanced A - Original	AUPRC	-0.026	-0.073	0.018	0.122
Balanced B - Original	AUROC	-0.006	-0.037	0.026	0.351
Balanced B - Original	AUPRC	-0.047	-0.099	-0.001	0.022
RF - Balanced B	AUROC	0.095	0.024	0.171	0.997
RF - Balanced B	AUPRC	0.146	0.073	0.213	1.000

11 Training audit and reproducibility

Table M7: All fits stopped before max_iter=2000; no recorded training warnings. This is not proof of global optimization.

Variant	FITS	Min iter	Max iter	WARNINGS
Balanced A	31	14	52	0
Balanced B	331	12	77	0
Original	331	12	62	0

Python 3.12.8; scikit-learn 1.9.0; NumPy 2.5.1; pandas 2.3.3. Model seed 20260824; inner seeds 20260825–20260829; bootstrap seed 20260824. Full-Development class weights: 0.7265306122 (negative), 1.6036036036 (toxic). All five outer and twenty inner CV splits are cluster disjoint. Native early-stopping reservations are not: 14–32 clusters overlap their gradient subsets. No resampling or evaluation weighting was used. All CSV outputs match an independent complete rerun exactly.

Repeat from the repository root with its historical environment:

```
.\.venv\Scripts\python.exe -m src.mlp_sensitivity
# Add --output-dir NEW_PATH; existing bundles are never overwritten.
.\.venv\Scripts\python.exe -m src.mlp_sensitivity_report
# Add --run NEW_PATH --output-dir NEW_REPORT_PATH
```

Authoritative bundle: `results/mlp_sensitivity/full_20261009_v2`.

The bundle contains input/source hashes, exact split memberships, candidate-level AUROCs, selected parameters, threshold OOF predictions, full fit/stopping histories, model artifacts, post-freeze external predictions and metrics, paired statistics, a sensitivity figure and integrity manifests. The integrated manuscript maps this additional analysis to Methods II-D/II-E and Results III-B/III-C/III-E without deleting the initial limitation.

Library references: Official MLPClassifier API documentation and the locally hashed 1.9.0 implementation. No runtime dependency was upgraded.